

[illegible]

Simplify deltas:

[illegible]

[illegible]

[illegible]

[illegible]

Simplify R:

[illegible]

$\xrightarrow{\text{cleaned up}} R_z^{-17} \cdot \frac{1}{11025} \cdot$

$$\begin{aligned}
& +2027025 \cdot R_z^8 \cdot \Phi_{zzzz}^B \cdot \Phi_{zzzz}^A - 135135 \cdot R_z^8 \cdot \Phi_{zzzz}^B \cdot \Phi_{zz\eta\eta}^A \\
& -270270 \cdot R_z^8 \cdot \Phi_{zzzz}^B \cdot \Phi_{zz\zeta\zeta}^A - 405405 \cdot R_z^8 \cdot \Phi_{zzzz}^B \cdot \Phi_{zz\epsilon\epsilon}^A \\
& -540540 \cdot R_z^8 \cdot \Phi_{zzz\delta}^B \cdot \Phi_{zzz\delta}^A - 540540 \cdot R_z^8 \cdot \Phi_{zzz\gamma}^B \cdot \Phi_{zzz\gamma}^A \\
& -135135 \cdot R_z^8 \cdot \Phi_{zz\gamma\gamma}^B \cdot \Phi_{zzzz}^A - 540540 \cdot R_z^8 \cdot \Phi_{zzz\beta}^B \cdot \Phi_{zzz\beta}^A \\
& -135135 \cdot R_z^8 \cdot \Phi_{zz\beta\beta}^B \cdot \Phi_{zzzz}^A - 135135 \cdot R_z^8 \cdot \Phi_{zz\beta\beta}^B \cdot \Phi_{zzzz}^A \\
& -540540 \cdot R_z^8 \cdot \Phi_{zzz\alpha}^B \cdot \Phi_{zzz\alpha}^A - 135135 \cdot R_z^8 \cdot \Phi_{zz\alpha\alpha}^B \cdot \Phi_{zzzz}^A \\
& -135135 \cdot R_z^8 \cdot \Phi_{zz\alpha\alpha}^B \cdot \Phi_{zzzz}^A - 135135 \cdot R_z^8 \cdot \Phi_{zz\alpha\alpha}^B \cdot \Phi_{zzzz}^A \\
& +31185 \cdot R_z^8 \cdot \Phi_{zzzz}^B \cdot \Phi_{\epsilon\epsilon\zeta\zeta}^A \\
& +31185 \cdot R_z^8 \cdot \Phi_{zzz\delta}^B \cdot \Phi_{z\delta\zeta\zeta}^A + 93555 \cdot R_z^8 \cdot \Phi_{zzz\delta}^B \cdot \Phi_{z\delta\epsilon\epsilon}^A \\
& +31185 \cdot R_z^8 \cdot \Phi_{zzz\gamma}^B \cdot \Phi_{z\gamma\zeta\zeta}^A + 93555 \cdot R_z^8 \cdot \Phi_{zzz\gamma}^B \cdot \Phi_{z\gamma\epsilon\epsilon}^A \\
& +62370 \cdot R_z^8 \cdot \Phi_{zz\gamma\gamma}^B \cdot \Phi_{zz\delta\delta}^A + 124740 \cdot R_z^8 \cdot \Phi_{zz\gamma\delta}^B \cdot \Phi_{zz\gamma\delta}^A \\
& +31185 \cdot R_z^8 \cdot \Phi_{zzz\beta}^B \cdot \Phi_{z\beta\zeta\zeta}^A + 93555 \cdot R_z^8 \cdot \Phi_{zzz\beta}^B \cdot \Phi_{z\beta\epsilon\epsilon}^A \\
& +62370 \cdot R_z^8 \cdot \Phi_{zz\beta\beta}^B \cdot \Phi_{zz\delta\delta}^A + 124740 \cdot R_z^8 \cdot \Phi_{zz\beta\delta}^B \cdot \Phi_{zz\beta\delta}^A \\
& +62370 \cdot R_z^8 \cdot \Phi_{zz\beta\beta}^B \cdot \Phi_{zz\gamma\gamma}^A + 124740 \cdot R_z^8 \cdot \Phi_{zz\beta\gamma}^B \cdot \Phi_{zz\beta\gamma}^A \\
& +83160 \cdot R_z^8 \cdot \Phi_{z\beta\beta\gamma}^B \cdot \Phi_{zzz\gamma}^A + 41580 \cdot R_z^8 \cdot \Phi_{z\beta\gamma\gamma}^B \cdot \Phi_{zzz\beta}^A \\
& +31185 \cdot R_z^8 \cdot \Phi_{zzz\alpha}^B \cdot \Phi_{z\alpha\zeta\zeta}^A + 93555 \cdot R_z^8 \cdot \Phi_{zzz\alpha}^B \cdot \Phi_{z\alpha\epsilon\epsilon}^A \\
& +62370 \cdot R_z^8 \cdot \Phi_{zz\alpha\alpha}^B \cdot \Phi_{zz\delta\delta}^A + 124740 \cdot R_z^8 \cdot \Phi_{zz\alpha\delta}^B \cdot \Phi_{zz\alpha\delta}^A \\
& +62370 \cdot R_z^8 \cdot \Phi_{zz\alpha\alpha}^B \cdot \Phi_{zz\gamma\gamma}^A + 124740 \cdot R_z^8 \cdot \Phi_{zz\alpha\gamma}^B \cdot \Phi_{zz\alpha\gamma}^A \\
& +83160 \cdot R_z^8 \cdot \Phi_{z\alpha\alpha\gamma}^B \cdot \Phi_{zzz\gamma}^A + 41580 \cdot R_z^8 \cdot \Phi_{z\alpha\gamma\gamma}^B \cdot \Phi_{zzz\alpha}^A \\
& +62370 \cdot R_z^8 \cdot \Phi_{zz\alpha\alpha}^B \cdot \Phi_{zz\beta\beta}^A + 124740 \cdot R_z^8 \cdot \Phi_{zz\alpha\beta}^B \cdot \Phi_{zz\alpha\beta}^A \\
& +83160 \cdot R_z^8 \cdot \Phi_{z\alpha\alpha\beta}^B \cdot \Phi_{zzz\beta}^A + 41580 \cdot R_z^8 \cdot \Phi_{z\alpha\beta\beta}^B \cdot \Phi_{zzz\alpha}^A \\
& +83160 \cdot R_z^8 \cdot \Phi_{z\alpha\alpha\beta}^B \cdot \Phi_{zzz\beta}^A + 41580 \cdot R_z^8 \cdot \Phi_{z\alpha\beta\beta}^B \cdot \Phi_{zzz\alpha}^A \\
& +31185 \cdot R_z^8 \cdot \Phi_{\alpha\alpha\beta\beta}^B \cdot \Phi_{zzzz}^A - 2835 \cdot R_z^8 \cdot \Phi_{zz\gamma\gamma}^B \cdot \Phi_{\delta\delta\epsilon\epsilon}^A \\
& -11340 \cdot R_z^8 \cdot \Phi_{zz\gamma\delta}^B \cdot \Phi_{\gamma\delta\epsilon\epsilon}^A \\
& -2835 \cdot R_z^8 \cdot \Phi_{zz\beta\beta}^B \cdot \Phi_{\delta\delta\epsilon\epsilon}^A \\
& -11340 \cdot R_z^8 \cdot \Phi_{zz\beta\delta}^B \cdot \Phi_{\beta\delta\epsilon\epsilon}^A \\
& -2835 \cdot R_z^8 \cdot \Phi_{zz\beta\beta}^B \cdot \Phi_{\gamma\gamma\epsilon\epsilon}^A \\
& -11340 \cdot R_z^8 \cdot \Phi_{zz\beta\gamma}^B \cdot \Phi_{\beta\gamma\epsilon\epsilon}^A \\
& -22680 \cdot R_z^8 \cdot \Phi_{z\beta\beta\gamma}^B \cdot \Phi_{z\gamma\delta\delta}^A \\
& -11340 \cdot R_z^8 \cdot \Phi_{z\beta\gamma\gamma}^B \cdot \Phi_{z\beta\delta\delta}^A \\
& -22680 \cdot R_z^8 \cdot \Phi_{z\beta\gamma\delta}^B \cdot \Phi_{z\beta\gamma\delta}^A \\
& -2835 \cdot R_z^8 \cdot \Phi_{zz\alpha\alpha}^B \cdot \Phi_{\delta\delta\epsilon\epsilon}^A \\
& -11340 \cdot R_z^8 \cdot \Phi_{zz\alpha\delta}^B \cdot \Phi_{\alpha\delta\epsilon\epsilon}^A \\
& -2835 \cdot R_z^8 \cdot \Phi_{zz\alpha\alpha}^B \cdot \Phi_{\gamma\gamma\epsilon\epsilon}^A \\
& -11340 \cdot R_z^8 \cdot \Phi_{zz\alpha\gamma}^B \cdot \Phi_{\alpha\gamma\epsilon\epsilon}^A \\
& -22680 \cdot R_z^8 \cdot \Phi_{z\alpha\alpha\gamma}^B \cdot \Phi_{z\gamma\delta\delta}^A \\
& -11340 \cdot R_z^8 \cdot \Phi_{z\alpha\gamma\gamma}^B \cdot \Phi_{z\alpha\delta\delta}^A \\
& -22680 \cdot R_z^8 \cdot \Phi_{z\alpha\gamma\delta}^B \cdot \Phi_{z\alpha\gamma\delta}^A \\
& -2835 \cdot R_z^8 \cdot \Phi_{zz\alpha\alpha}^B \cdot \Phi_{\beta\beta\epsilon\epsilon}^A \\
& -11340 \cdot R_z^8 \cdot \Phi_{zz\alpha\beta}^B \cdot \Phi_{\alpha\beta\epsilon\epsilon}^A \\
& -22680 \cdot R_z^8 \cdot \Phi_{z\alpha\alpha\beta}^B \cdot \Phi_{z\beta\delta\delta}^A \\
& -11340 \cdot R_z^8 \cdot \Phi_{z\alpha\beta\beta}^B \cdot \Phi_{z\alpha\delta\delta}^A \\
& -22680 \cdot R_z^8 \cdot \Phi_{z\alpha\beta\delta}^B \cdot \Phi_{z\alpha\beta\delta}^A \\
& -22680 \cdot R_z^8 \cdot \Phi_{z\alpha\alpha\beta}^B \cdot \Phi_{z\beta\gamma\gamma}^A \\
& -11340 \cdot R_z^8 \cdot \Phi_{z\alpha\beta\beta}^B \cdot \Phi_{z\alpha\gamma\gamma}^A \\
& -22680 \cdot R_z^8 \cdot \Phi_{z\alpha\beta\gamma}^B \cdot \Phi_{z\alpha\beta\gamma}^A \\
& -17010 \cdot R_z^8 \cdot \Phi_{\alpha\alpha\beta\beta}^B \cdot \Phi_{zz\gamma\gamma}^A \\
& -34020 \cdot R_z^8 \cdot \Phi_{\alpha\alpha\beta\gamma}^B \cdot \Phi_{zz\beta\gamma}^A \\
& -22680 \cdot R_z^8 \cdot \Phi_{\alpha\beta\beta\gamma}^B \cdot \Phi_{zz\alpha\gamma}^A \\
& -11340 \cdot R_z^8 \cdot \Phi_{\alpha\beta\gamma\gamma}^B \cdot \Phi_{zz\alpha\beta}^A \\
& 9+945 \cdot R_z^8 \cdot \Phi_{\alpha\alpha\beta\beta}^B \cdot \Phi_{\gamma\gamma\delta\delta}^A \\
& +3780 \cdot R_z^8 \cdot \Phi_{\alpha\alpha\beta\gamma}^B \cdot \Phi_{\beta\gamma\delta\delta}^A \\
& +2520 \cdot R_z^8 \cdot \Phi_{\alpha\beta\beta\gamma}^B \cdot \Phi_{\gamma\gamma\delta\delta}^A
\end{aligned}$$

(3b)

[illegible]

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$$\begin{aligned}
& \xrightarrow{\text{added up}} R_z^{-17} \cdot \frac{1}{11025} \cdot \left[\begin{aligned}
& +2027025 \cdot R_z^8 \cdot \Phi_{zzzz}^B \cdot \Phi_{zzzz}^A - 810810 \cdot R_z^8 \cdot \Phi_{zzzz}^B \cdot \Phi_{zz\alpha\alpha}^A \\
& -2162160 \cdot R_z^8 \cdot \Phi_{zzz\alpha}^B \cdot \Phi_{zzz\alpha}^A - 810810 \cdot R_z^8 \cdot \Phi_{zzz\alpha}^B \cdot \Phi_{zzzz}^A \\
& +31185 \cdot R_z^8 \cdot \Phi_{zzzz}^B \cdot \Phi_{\alpha\alpha\beta\beta}^A + 374220 \cdot R_z^8 \cdot \Phi_{zzzz}^B \cdot \Phi_{z\alpha\beta\beta}^A \\
& +249480 \cdot R_z^8 \cdot \Phi_{zz\alpha\alpha}^B \cdot \Phi_{zz\beta\beta}^A + 748440 \cdot R_z^8 \cdot \Phi_{zz\alpha\beta}^B \cdot \Phi_{zz\alpha\beta}^A \\
& +124740 \cdot R_z^8 \cdot \Phi_{zzz\beta}^B \cdot \Phi_{z\alpha\alpha\beta}^A + 124740 \cdot R_z^8 \cdot \Phi_{zz\beta\beta}^B \cdot \Phi_{zz\alpha\alpha}^A \\
& +207900 \cdot R_z^8 \cdot \Phi_{z\alpha\beta\beta}^B \cdot \Phi_{zzz\alpha}^A + 291060 \cdot R_z^8 \cdot \Phi_{z\alpha\alpha\beta}^B \cdot \Phi_{zzz\beta}^A \\
& +31185 \cdot R_z^8 \cdot \Phi_{\alpha\alpha\beta\beta}^B \cdot \Phi_{zzzz}^A - 2835 \cdot R_z^8 \cdot \Phi_{zz\gamma\gamma}^B \cdot \Phi_{\alpha\alpha\beta\beta}^A \\
& -22680 \cdot R_z^8 \cdot \Phi_{zz\alpha\gamma}^B \cdot \Phi_{\alpha\beta\beta\gamma}^A \\
& -5670 \cdot R_z^8 \cdot \Phi_{zz\beta\beta}^B \cdot \Phi_{\alpha\alpha\gamma\gamma}^A \\
& -34020 \cdot R_z^8 \cdot \Phi_{zz\alpha\beta}^B \cdot \Phi_{\alpha\beta\gamma\gamma}^A \\
& -11340 \cdot R_z^8 \cdot \Phi_{zz\beta\gamma}^B \cdot \Phi_{\alpha\alpha\beta\gamma}^A \\
& -22680 \cdot R_z^8 \cdot \Phi_{z\beta\beta\gamma}^B \cdot \Phi_{z\alpha\alpha\gamma}^A \\
& -11340 \cdot R_z^8 \cdot \Phi_{z\beta\gamma\gamma}^B \cdot \Phi_{z\alpha\alpha\beta}^A \\
& -90720 \cdot R_z^8 \cdot \Phi_{z\alpha\beta\gamma}^B \cdot \Phi_{z\alpha\beta\gamma}^A \\
& -8505 \cdot R_z^8 \cdot \Phi_{zz\alpha\alpha}^B \cdot \Phi_{\beta\beta\gamma\gamma}^A \\
& -22680 \cdot R_z^8 \cdot \Phi_{z\alpha\alpha\gamma}^B \cdot \Phi_{z\beta\beta\gamma}^A \\
& -11340 \cdot R_z^8 \cdot \Phi_{z\alpha\gamma\gamma}^B \cdot \Phi_{z\alpha\beta\beta}^A \\
& -45360 \cdot R_z^8 \cdot \Phi_{z\alpha\alpha\beta}^B \cdot \Phi_{z\beta\gamma\gamma}^A \\
& -22680 \cdot R_z^8 \cdot \Phi_{z\alpha\beta\beta}^B \cdot \Phi_{z\alpha\gamma\gamma}^A \\
& -17010 \cdot R_z^8 \cdot \Phi_{\alpha\alpha\beta\beta}^B \cdot \Phi_{zz\gamma\gamma}^A \\
& -34020 \cdot R_z^8 \cdot \Phi_{\alpha\alpha\beta\gamma}^B \cdot \Phi_{zz\beta\gamma}^A \\
& -22680 \cdot R_z^8 \cdot \Phi_{\alpha\beta\beta\gamma}^B \cdot \Phi_{zz\alpha\gamma}^A \\
& -11340 \cdot R_z^8 \cdot \Phi_{\alpha\beta\gamma\gamma}^B \cdot \Phi_{zz\alpha\beta}^A \\
& +945 \cdot R_z^8 \cdot \Phi_{\alpha\alpha\beta\beta}^B \cdot \Phi_{\gamma\gamma\delta\delta}^A \\
& +3780 \cdot R_z^8 \cdot \Phi_{\alpha\alpha\beta\gamma}^B \cdot \Phi_{\beta\gamma\delta\delta}^A \\
& +2520 \cdot R_z^8 \cdot \Phi_{\alpha\beta\beta\gamma}^B \cdot \Phi_{\alpha\gamma\delta\delta}^A \\
& +1260 \cdot R_z^8 \cdot \Phi_{\alpha\beta\gamma\gamma}^B \cdot \Phi_{\alpha\beta\delta\delta}^A \\
& +2520 \cdot R_z^8 \cdot \Phi_{\alpha\beta\gamma\delta}^B \cdot \Phi_{\alpha\beta\gamma\delta}^A
\end{aligned} \right] \tag{3d}
\end{aligned}$$

Simplify Hexadecapoles:

$$\begin{aligned}
& + \frac{1}{11025} \cdot T_{\alpha\beta\gamma\delta\epsilon\zeta\eta\theta} \Phi_{\alpha\beta\gamma\delta}^B \Phi_{\epsilon\zeta\eta\theta}^A = R_z^{-17} \cdot \frac{1}{11025} \cdot \left[\begin{aligned}
& +2027025 \cdot R_z^8 \cdot \Phi_{zzzz}^B \cdot \Phi_{zzzz}^A - 810810 \cdot R_z^8 \cdot \Phi_{zzzz}^B \cdot \Phi_{zz\alpha\alpha}^A \\
& -2162160 \cdot R_z^8 \cdot \Phi_{zzz\alpha}^B \cdot \Phi_{zzz\alpha}^A - 810810 \cdot R_z^8 \cdot \Phi_{zz\alpha\alpha}^B \cdot \Phi_{zzzz}^A \\
& +31185 \cdot R_z^8 \cdot \Phi_{zzzz}^B \cdot \Phi_{\alpha\alpha\beta\beta}^A + 374220 \cdot R_z^8 \cdot \Phi_{zzzz}^B \cdot \Phi_{zz\beta\beta}^A \\
& +249480 \cdot R_z^8 \cdot \Phi_{zz\alpha\alpha}^B \cdot \Phi_{zz\beta\beta}^A + 748440 \cdot R_z^8 \cdot \Phi_{zz\alpha\alpha}^B \cdot \Phi_{zz\alpha\alpha}^A \\
& +124740 \cdot R_z^8 \cdot \Phi_{zzzz}^B \cdot \Phi_{zz\alpha\alpha}^A + 124740 \cdot R_z^8 \cdot \Phi_{zz\beta\beta}^B \cdot \Phi_{zz\alpha\alpha}^A \\
& +207900 \cdot R_z^8 \cdot \Phi_{zz\beta\beta}^B \cdot \Phi_{zzzz}^A + 291060 \cdot R_z^8 \cdot \Phi_{zz\alpha\alpha}^B \cdot \Phi_{zzzz}^A \\
& +31185 \cdot R_z^8 \cdot \Phi_{\alpha\alpha\beta\beta}^B \cdot \Phi_{zzzz}^A - 2835 \cdot R_z^8 \cdot \Phi_{zz\gamma\gamma}^B \cdot \Phi_{\alpha\alpha\beta\beta}^A \\
& -22680 \cdot R_z^8 \cdot \Phi_{zz\alpha\alpha}^B \cdot \Phi_{\alpha\alpha\beta\beta}^A \\
& -5670 \cdot R_z^8 \cdot \Phi_{zz\beta\beta}^B \cdot \Phi_{\alpha\alpha\gamma\gamma}^A \\
& -34020 \cdot R_z^8 \cdot \Phi_{zz\alpha\alpha}^B \cdot \Phi_{\alpha\alpha\gamma\gamma}^A \\
& -11340 \cdot R_z^8 \cdot \Phi_{zz\beta\beta}^B \cdot \Phi_{\alpha\alpha\beta\beta}^A \\
& -22680 \cdot R_z^8 \cdot \Phi_{zz\beta\beta}^B \cdot \Phi_{zz\alpha\alpha}^A - 11340 \cdot R_z^8 \cdot \Phi_{zz\gamma\gamma}^B \cdot \Phi_{zz\alpha\alpha}^A \\
& -90720 \cdot R_z^8 \cdot \Phi_{zz\alpha\beta\gamma}^B \cdot \Phi_{z\alpha\beta\gamma}^A \\
& -8505 \cdot R_z^8 \cdot \Phi_{zz\alpha\alpha}^B \cdot \Phi_{\beta\beta\gamma\gamma}^A \\
& -22680 \cdot R_z^8 \cdot \Phi_{zz\alpha\alpha}^B \cdot \Phi_{zz\beta\beta}^A - 11340 \cdot R_z^8 \cdot \Phi_{zz\gamma\gamma}^B \cdot \Phi_{zz\beta\beta}^A \\
& -45360 \cdot R_z^8 \cdot \Phi_{zz\alpha\alpha}^B \cdot \Phi_{zz\gamma\gamma}^A - 22680 \cdot R_z^8 \cdot \Phi_{zz\beta\beta}^B \cdot \Phi_{zz\gamma\gamma}^A \\
& -17010 \cdot R_z^8 \cdot \Phi_{\alpha\alpha\beta\beta}^B \cdot \Phi_{zz\gamma\gamma}^A \\
& -34020 \cdot R_z^8 \cdot \Phi_{\alpha\alpha\beta\beta}^B \cdot \Phi_{zz\beta\beta}^A \\
& -22680 \cdot R_z^8 \cdot \Phi_{\alpha\alpha\beta\beta}^B \cdot \Phi_{zz\alpha\alpha}^A \\
& -11340 \cdot R_z^8 \cdot \Phi_{\alpha\alpha\gamma\gamma}^B \cdot \Phi_{zz\alpha\alpha}^A \\
& +945 \cdot R_z^8 \cdot \Phi_{\alpha\alpha\beta\beta}^B \cdot \Phi_{\gamma\gamma\delta\delta}^A \\
& +3780 \cdot R_z^8 \cdot \Phi_{\alpha\alpha\beta\beta}^B \cdot \Phi_{\beta\beta\delta\delta}^A \\
& +2520 \cdot R_z^8 \cdot \Phi_{\alpha\alpha\beta\beta}^B \cdot \Phi_{\alpha\alpha\delta\delta}^A \\
& +1260 \cdot R_z^8 \cdot \Phi_{\alpha\alpha\gamma\gamma}^B \cdot \Phi_{\alpha\alpha\delta\delta}^A \\
& +2520 \cdot R_z^8 \cdot \Phi_{\alpha\beta\gamma\delta}^B \cdot \Phi_{\alpha\beta\gamma\delta}^A
\end{aligned} \right]
\end{aligned} \tag{4a}$$

$$\begin{aligned}
& \xrightarrow{\text{reduced indices}} R_z^{-17} \cdot \frac{1}{11025} \cdot \left[\begin{aligned}
& +2027025 \cdot R_z^8 \cdot \Phi_{zzzz}^B \cdot \Phi_{zzzz}^A - 810810 \cdot R_z^8 \cdot \Phi_{zzzz}^B \cdot \Phi_{zz\alpha\alpha}^A \\
& -2162160 \cdot R_z^8 \cdot \Phi_{zz\alpha\alpha}^B \cdot \Phi_{zz\alpha\alpha}^A - 810810 \cdot R_z^8 \cdot \Phi_{zz\alpha\alpha}^B \cdot \Phi_{zzzz}^A \\
& +31185 \cdot R_z^8 \cdot \Phi_{zzzz}^B \cdot \Phi_{\alpha\alpha\beta\beta}^A + 374220 \cdot R_z^8 \cdot \Phi_{zzzz}^B \cdot \Phi_{zz\alpha\alpha}^A \\
& +249480 \cdot R_z^8 \cdot \Phi_{zz\alpha\alpha}^B \cdot \Phi_{zz\beta\beta}^A + 748440 \cdot R_z^8 \cdot \Phi_{zz\alpha\alpha}^B \cdot \Phi_{zz\alpha\alpha}^A \\
& +124740 \cdot R_z^8 \cdot \Phi_{zzzz}^B \cdot \Phi_{zz\alpha\alpha}^A + 124740 \cdot R_z^8 \cdot \Phi_{zz\alpha\alpha}^B \cdot \Phi_{zz\alpha\alpha}^A \\
& +207900 \cdot R_z^8 \cdot \Phi_{zz\alpha\alpha}^B \cdot \Phi_{zzzz}^A + 291060 \cdot R_z^8 \cdot \Phi_{zz\alpha\alpha}^B \cdot \Phi_{zzzz}^A \\
& +31185 \cdot R_z^8 \cdot \Phi_{\alpha\alpha\beta\beta}^B \cdot \Phi_{zzzz}^A - 2835 \cdot R_z^8 \cdot \Phi_{zz\alpha\alpha}^B \cdot \Phi_{\beta\beta\gamma\gamma}^A \\
& -22680 \cdot R_z^8 \cdot \Phi_{zz\alpha\alpha}^B \cdot \Phi_{\alpha\alpha\beta\beta}^A \\
& -5670 \cdot R_z^8 \cdot \Phi_{zz\beta\beta}^B \cdot \Phi_{\alpha\alpha\gamma\gamma}^A \\
& -34020 \cdot R_z^8 \cdot \Phi_{zz\alpha\alpha}^B \cdot \Phi_{\alpha\alpha\beta\beta}^A \\
& -11340 \cdot R_z^8 \cdot \Phi_{zz\alpha\alpha}^B \cdot \Phi_{\alpha\alpha\beta\beta}^A \\
& -22680 \cdot R_z^8 \cdot \Phi_{zz\alpha\alpha}^B \cdot \Phi_{zz\beta\beta}^A - 11340 \cdot R_z^8 \cdot \Phi_{zz\alpha\alpha}^B \cdot \Phi_{zz\beta\beta}^A \\
& -90720 \cdot R_z^8 \cdot \Phi_{zz\alpha\alpha}^B \cdot \Phi_{z\alpha\beta\gamma}^A \\
& -8505 \cdot R_z^8 \cdot \Phi_{zz\alpha\alpha}^B \cdot \Phi_{\beta\beta\gamma\gamma}^A \\
& -22680 \cdot R_z^8 \cdot \Phi_{zz\alpha\alpha}^B \cdot \Phi_{zz\beta\beta}^A - 11340 \cdot R_z^8 \cdot \Phi_{zz\alpha\alpha}^B \cdot \Phi_{zz\beta\beta}^A \\
& -45360 \cdot R_z^8 \cdot \Phi_{zz\alpha\alpha}^B \cdot \Phi_{zz\beta\beta}^A - 22680 \cdot R_z^8 \cdot \Phi_{zz\alpha\alpha}^B \cdot \Phi_{zz\beta\beta}^A \\
& -17010 \cdot R_z^8 \cdot \Phi_{\alpha\alpha\beta\beta}^B \cdot \Phi_{zz\gamma\gamma}^A \\
& -34020 \cdot R_z^8 \cdot \Phi_{\alpha\alpha\beta\beta}^B \cdot \Phi_{zz\beta\beta}^A \\
& -22680 \cdot R_z^8 \cdot \Phi_{\alpha\alpha\beta\beta}^B \cdot \Phi_{zz\alpha\alpha}^A \\
& -11340 \cdot R_z^8 \cdot \Phi_{\alpha\alpha\beta\beta}^B \cdot \Phi_{zz\alpha\alpha}^A \\
& +945 \cdot R_z^8 \cdot \Phi_{\alpha\alpha\beta\beta}^B \cdot \Phi_{\gamma\gamma\delta\delta}^A \\
& +3780 \cdot R_z^8 \cdot \Phi_{\alpha\alpha\beta\beta}^B \cdot \Phi_{\beta\beta\gamma\gamma}^A \\
& +2520 \cdot R_z^8 \cdot \Phi_{\alpha\alpha\beta\beta}^B \cdot \Phi_{\alpha\alpha\gamma\gamma}^A \\
& +1260 \cdot R_z^8 \cdot \Phi_{\alpha\alpha\gamma\gamma}^B \cdot \Phi_{\beta\beta\gamma\gamma}^A \\
& +2520 \cdot R_z^8 \cdot \Phi_{\alpha\beta\gamma\delta}^B \cdot \Phi_{\alpha\beta\gamma\delta}^A
\end{aligned} \right] \quad (4b)
\end{aligned}$$

$$\begin{aligned}
& \xrightarrow{\text{added up}} R_z^{-17} \cdot \frac{1}{11025} \cdot \left[\begin{aligned}
& +2027025 \cdot R_z^8 \cdot \Phi_{zzzz}^B \cdot \Phi_{zzzz}^A - 311850 \cdot R_z^8 \cdot \Phi_{zzzz}^B \cdot \Phi_{zz\alpha\alpha}^A \\
& -2162160 \cdot R_z^8 \cdot \Phi_{zz\alpha\alpha}^B \cdot \Phi_{zz\alpha\alpha}^A - 311850 \cdot R_z^8 \cdot \Phi_{zz\alpha\alpha}^B \cdot \Phi_{zzzz}^A \\
& +31185 \cdot R_z^8 \cdot \Phi_{zzzz}^B \cdot \Phi_{\alpha\alpha\beta\beta}^A + 238140 \cdot R_z^8 \cdot \Phi_{zz\alpha\alpha}^B \cdot \Phi_{zz\beta\beta}^A \\
& +748440 \cdot R_z^8 \cdot \Phi_{zz\alpha\alpha}^B \cdot \Phi_{zz\alpha\alpha}^A \\
& +31185 \cdot R_z^8 \cdot \Phi_{\alpha\alpha\beta\beta}^B \cdot \Phi_{zzzz}^A - 11340 \cdot R_z^8 \cdot \Phi_{zz\alpha\alpha}^B \cdot \Phi_{\beta\beta\gamma\gamma}^A \\
& -68040 \cdot R_z^8 \cdot \Phi_{zz\alpha\alpha}^B \cdot \Phi_{\alpha\alpha\beta\beta}^A \\
& -5670 \cdot R_z^8 \cdot \Phi_{zz\beta\beta}^B \cdot \Phi_{\alpha\alpha\gamma\gamma}^A \\
& -90720 \cdot R_z^8 \cdot \Phi_{zz\alpha\alpha}^B \cdot \Phi_{z\alpha\beta\gamma}^A \\
& -17010 \cdot R_z^8 \cdot \Phi_{\alpha\alpha\beta\beta}^B \cdot \Phi_{zz\gamma\gamma}^A \\
& -34020 \cdot R_z^8 \cdot \Phi_{\alpha\alpha\beta\beta}^B \cdot \Phi_{zz\beta\beta}^A \\
& -34020 \cdot R_z^8 \cdot \Phi_{\alpha\alpha\beta\beta}^B \cdot \Phi_{zz\alpha\alpha}^A \\
& +945 \cdot R_z^8 \cdot \Phi_{\alpha\alpha\beta\beta}^B \cdot \Phi_{\gamma\gamma\delta\delta}^A \\
& +3780 \cdot R_z^8 \cdot \Phi_{\alpha\alpha\beta\beta}^B \cdot \Phi_{\beta\beta\gamma\gamma}^A \\
& +2520 \cdot R_z^8 \cdot \Phi_{\alpha\alpha\beta\beta}^B \cdot \Phi_{\alpha\alpha\gamma\gamma}^A \\
& +1260 \cdot R_z^8 \cdot \Phi_{\alpha\alpha\gamma\gamma}^B \cdot \Phi_{\beta\beta\gamma\gamma}^A \\
& +2520 \cdot R_z^8 \cdot \Phi_{\alpha\beta\gamma\delta}^B \cdot \Phi_{\alpha\beta\gamma\delta}^A
\end{aligned} \right] \quad (4c)
\end{aligned}$$

Exploit Traceless Conditions:

$$\begin{aligned}
& + \frac{1}{11025} \cdot T_{\alpha\beta\gamma\delta\epsilon\zeta\eta\theta} \Phi_{\alpha\beta\gamma\delta}^B \Phi_{\epsilon\zeta\eta\theta}^A = R_z^{-17} \cdot \frac{1}{11025} \cdot \left[\begin{aligned}
& +2027025 \cdot R_z^8 \cdot \Phi_{zzzz}^B \cdot \Phi_{zzzz}^A - 2162160 \cdot R_z^8 \cdot \Phi_{zz\alpha\alpha}^B \cdot \Phi_{zz\alpha\alpha}^A \\
& +748440 \cdot R_z^8 \cdot \Phi_{zz\alpha\alpha}^B \cdot \Phi_{zz\alpha\alpha}^A \\
& -90720 \cdot R_z^8 \cdot \Phi_{zz\alpha\alpha}^B \cdot \Phi_{z\alpha\beta\gamma}^A \\
& +2520 \cdot R_z^8 \cdot \Phi_{\alpha\beta\gamma\delta}^B \cdot \Phi_{\alpha\beta\gamma\delta}^A
\end{aligned} \right] \quad (5)
\end{aligned}$$

Multiply Out Sum:

$$+\frac{1}{11025} \cdot T_{\alpha\beta\gamma\delta\epsilon\zeta\eta\theta} \Phi_{\alpha\beta\gamma\delta}^B \Phi_{\epsilon\zeta\eta\theta}^A = R_z^{-17} \cdot \frac{1}{11025} \cdot$$

[illegible]

$$\begin{aligned}
& \xrightarrow{\text{cleaned up}} R_z^{-17} \cdot \frac{1}{11025} \cdot \left[\begin{aligned}
& +2027025 \cdot R_z^8 \cdot \Phi_{zzzz}^B \cdot \Phi_{zzzz}^A - 2162160 \cdot R_z^8 \cdot \Phi_{zzzz}^B \cdot \Phi_{zzzz}^A \\
& +748440 \cdot R_z^8 \cdot \Phi_{xxzz}^B \cdot \Phi_{xxzz}^A + 748440 \cdot R_z^8 \cdot \Phi_{yyzz}^B \cdot \Phi_{yyzz}^A \\
& +748440 \cdot R_z^8 \cdot \Phi_{zzzz}^B \cdot \Phi_{zzzz}^A - 90720 \cdot R_z^8 \cdot \Phi_{xxzz}^B \cdot \Phi_{xxzz}^A \\
& -90720 \cdot R_z^8 \cdot \Phi_{xxzz}^B \cdot \Phi_{xxzz}^A - 90720 \cdot R_z^8 \cdot \Phi_{yyzz}^B \cdot \Phi_{yyzz}^A \\
& -90720 \cdot R_z^8 \cdot \Phi_{yyzz}^B \cdot \Phi_{yyzz}^A - 90720 \cdot R_z^8 \cdot \Phi_{xxzz}^B \cdot \Phi_{xxzz}^A \\
& -90720 \cdot R_z^8 \cdot \Phi_{yyzz}^B \cdot \Phi_{yyzz}^A - 90720 \cdot R_z^8 \cdot \Phi_{zzzz}^B \cdot \Phi_{zzzz}^A \\
& +2520 \cdot R_z^8 \cdot \Phi_{xxxx}^B \cdot \Phi_{xxxx}^A + 2520 \cdot R_z^8 \cdot \Phi_{xxyy}^B \cdot \Phi_{xxyy}^A \\
& +2520 \cdot R_z^8 \cdot \Phi_{xxzz}^B \cdot \Phi_{xxzz}^A + 2520 \cdot R_z^8 \cdot \Phi_{xxyy}^B \cdot \Phi_{xxyy}^A \\
& +2520 \cdot R_z^8 \cdot \Phi_{xxyy}^B \cdot \Phi_{xxyy}^A + 2520 \cdot R_z^8 \cdot \Phi_{xxzz}^B \cdot \Phi_{xxzz}^A \\
& +2520 \cdot R_z^8 \cdot \Phi_{xxyy}^B \cdot \Phi_{xxyy}^A + 2520 \cdot R_z^8 \cdot \Phi_{xxzz}^B \cdot \Phi_{xxzz}^A \\
& +2520 \cdot R_z^8 \cdot \Phi_{yyzz}^B \cdot \Phi_{yyzz}^A + 2520 \cdot R_z^8 \cdot \Phi_{yyzz}^B \cdot \Phi_{yyzz}^A \\
& +2520 \cdot R_z^8 \cdot \Phi_{yyzz}^B \cdot \Phi_{yyzz}^A + 2520 \cdot R_z^8 \cdot \Phi_{yyzz}^B \cdot \Phi_{yyzz}^A \\
& +2520 \cdot R_z^8 \cdot \Phi_{xxzz}^B \cdot \Phi_{xxzz}^A + 2520 \cdot R_z^8 \cdot \Phi_{xxzz}^B \cdot \Phi_{xxzz}^A \\
& +2520 \cdot R_z^8 \cdot \Phi_{yyzz}^B \cdot \Phi_{yyzz}^A + 2520 \cdot R_z^8 \cdot \Phi_{yyzz}^B \cdot \Phi_{yyzz}^A \\
& +2520 \cdot R_z^8 \cdot \Phi_{xxzz}^B \cdot \Phi_{xxzz}^A + 2520 \cdot R_z^8 \cdot \Phi_{yyzz}^B \cdot \Phi_{yyzz}^A \\
& +2520 \cdot R_z^8 \cdot \Phi_{xxzz}^B \cdot \Phi_{xxzz}^A + 2520 \cdot R_z^8 \cdot \Phi_{yyzz}^B \cdot \Phi_{yyzz}^A \\
& +2520 \cdot R_z^8 \cdot \Phi_{zzzz}^B \cdot \Phi_{zzzz}^A
\end{aligned} \right] \quad (6b)
\end{aligned}$$

$$\begin{aligned}
& \xrightarrow{\text{added up}} R_z^{-17} \cdot \frac{1}{11025} \cdot \left[\begin{aligned}
& +525105 \cdot R_z^8 \cdot \Phi_{zzzz}^B \cdot \Phi_{zzzz}^A + 491400 \cdot R_z^8 \cdot \Phi_{xxzz}^B \cdot \Phi_{xxzz}^A \\
& +491400 \cdot R_z^8 \cdot \Phi_{yyzz}^B \cdot \Phi_{yyzz}^A + 2520 \cdot R_z^8 \cdot \Phi_{xxxx}^B \cdot \Phi_{xxxx}^A \\
& +15120 \cdot R_z^8 \cdot \Phi_{xxyy}^B \cdot \Phi_{xxyy}^A + 2520 \cdot R_z^8 \cdot \Phi_{yyyy}^B \cdot \Phi_{yyyy}^A
\end{aligned} \right] \quad (6c)
\end{aligned}$$

Combining everything:

$$\begin{aligned}
& + \frac{1}{11025} \cdot T_{\alpha\beta\gamma\delta\epsilon\zeta\eta\theta} \Phi_{\alpha\beta\gamma\delta}^B \Phi_{\epsilon\zeta\eta\theta}^A = \left[\begin{aligned}
& + \frac{1667}{35} \cdot R_z^{-9} \cdot \Phi_{zzzz}^B \cdot \Phi_{zzzz}^A + \frac{312}{7} \cdot R_z^{-9} \cdot \Phi_{xxzz}^B \cdot \Phi_{xxzz}^A \\
& + \frac{312}{7} \cdot R_z^{-9} \cdot \Phi_{yyzz}^B \cdot \Phi_{yyzz}^A + \frac{8}{35} \cdot R_z^{-9} \cdot \Phi_{xxxx}^B \cdot \Phi_{xxxx}^A \\
& + \frac{48}{35} \cdot R_z^{-9} \cdot \Phi_{xxyy}^B \cdot \Phi_{xxyy}^A + \frac{8}{35} \cdot R_z^{-9} \cdot \Phi_{yyyy}^B \cdot \Phi_{yyyy}^A
\end{aligned} \right] \quad (7)
\end{aligned}$$